BANGALORE TECHNOLOGICAL INSTITUTE

(An ISO 9001:2015 Certified Institution)

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DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

(BCS701) INTERNET OF THINGS LAB MANUAL

SEMESTER –VII-(2025–26)

PREPARED BY:

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BANGALORE TECHNOLOGICAL INSTITUTE

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Kodathi Village,VarthoorHobli, Bangalore East Tq, Bangalore

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6	Develop a program to classify dry and wet waste with the Moisture sensor (DHT22)
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Step-by-Step: Tinkercad Arduino Installation and Setup

Step 1: Open Tinkercad Website

- Go to https://www.tinkercad.com in your browser.

Step 2: Create an Account

- Click **Join Now** or **Sign Up**.
- Fill out your details or sign in using Google/Microsoft accounts.
- Verify your email if required.

Step 3: Log in to Your Tinkercad Account

- Enter your credentials and log in.

Step 4: Navigate to Circuits

- Click the **Circuits** option on the left sidebar.

Step 5: Create a New Circuit

- Click on **Create new Circuit** button.

Step 6: Add an Arduino Board

- In the components panel on the right, search for **Arduino Uno**.
- Drag and drop it into the workspace.

Step 7: Add Components

- Drag other components like LEDs, resistors, sensors, buttons as needed.

Step 8: Write Code

- Click the **Code** button to open the code editor.
- You can use block-based or text-based Arduino code.

Step 9: Simulate Your Circuit

- Click **Start Simulation** to run your circuit and check functionality.

Step 10: Save Your Work

- Use the **Save** button to save your circuit project.

Tips:

- Always test your circuits in simulation before physical implementation.
- Use the Tinkercad learning tutorials for more help.
- Experiment with simple circuits first to get comfortable.

Program 1:

Develop a program to blink 5 LEDs back and forth.

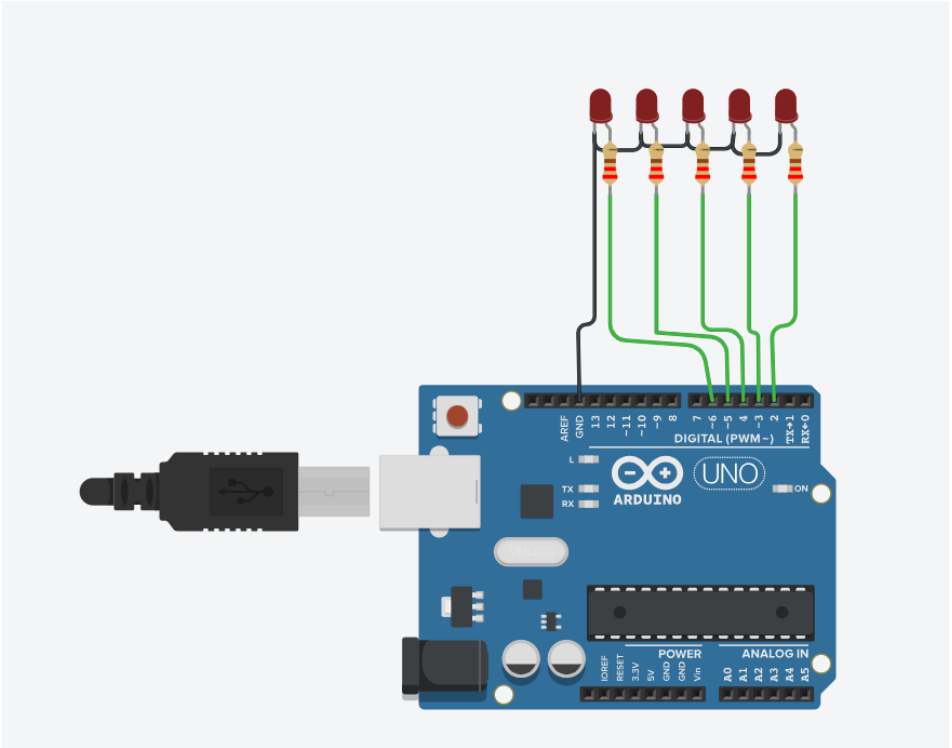
Aim:

To develop an Arduino program that makes 5 LEDs blink sequentially back and forth, creating a back-and-forth running light effect.

Component:

Name	Quantity	Component
U1	1	Arduino Uno R3
D1 D2 D3 D4 D5	5	Red LED
R1 R2 R3 R4 R5	5	220 Ω Resistor

Circuit:



Procedure:

Step 1: Open Tinkercad

1. Go to https://www.tinkercad.com
2. Click **"Circuits" → "Create new Circuit"**

Step 2: Add Components

- Drag and drop:
  - **1 Arduino Uno**
  - **5 LEDs**
  - **5 Resistors**
  - (Optional) **1 Breadboard**

Step 3: Wiring the LEDs

For each LED:

- **Anode (long leg)** → connect to a **digital pin** via a **resistor (220 Ω)**:
  - LED1 → Pin 2
  - LED2 → Pin 3
  - LED3 → Pin 4
  - LED4 → Pin 5
  - LED5 → Pin 6
- **Cathode (short leg)** → connect to **GND**

Step 4: Add the Arduino Code

Click **"Code"** → change from **Blocks** to **Text**.
Delete the default code and write the following:

Program Code:

```
int ledPins[] = {2, 3, 4, 5, 6};
int numLeds = 5;
int delayTime = 150;

void setup() {
  for (int i = 0; i < numLeds; i++) {
    pinMode(ledPins[i], OUTPUT);
  }
}

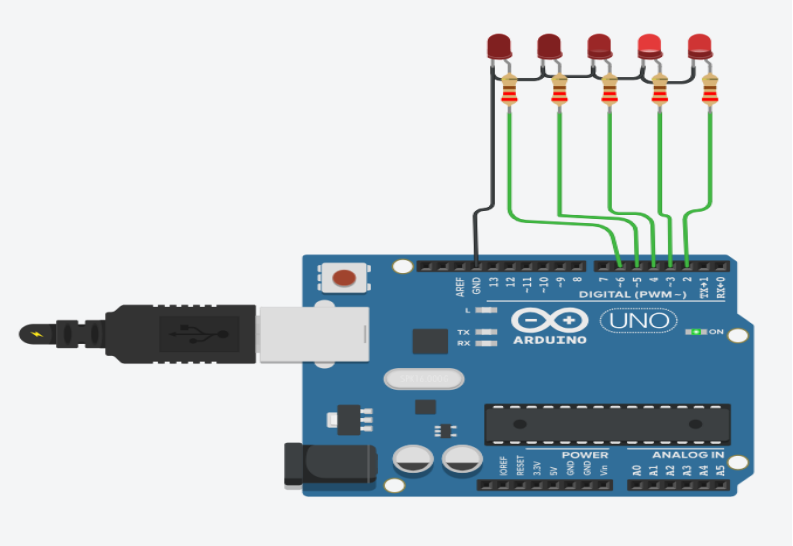
void loop() {
  for (int i = 0; i < numLeds; i++) {
    turnOnOnly(i);
    delay(delayTime);
  }
```

```
  for (int i = numLeds - 2; i > 0; i--) {
    turnOnOnly(i);
    delay(delayTime);
  }
}
void turnOnOnly(int idx) {
  for (int j = 0; j < numLeds; j++) {
    digitalWrite(ledPins[j], (j == idx));
  }
}
```

Step 5: Start Simulation

- Click **"Start Simulation"**
- LEDs will blink **from left to right**, then **right to left**, continuously

Output:



Program 2:
Develop a program to interface a relay with Arduino board.

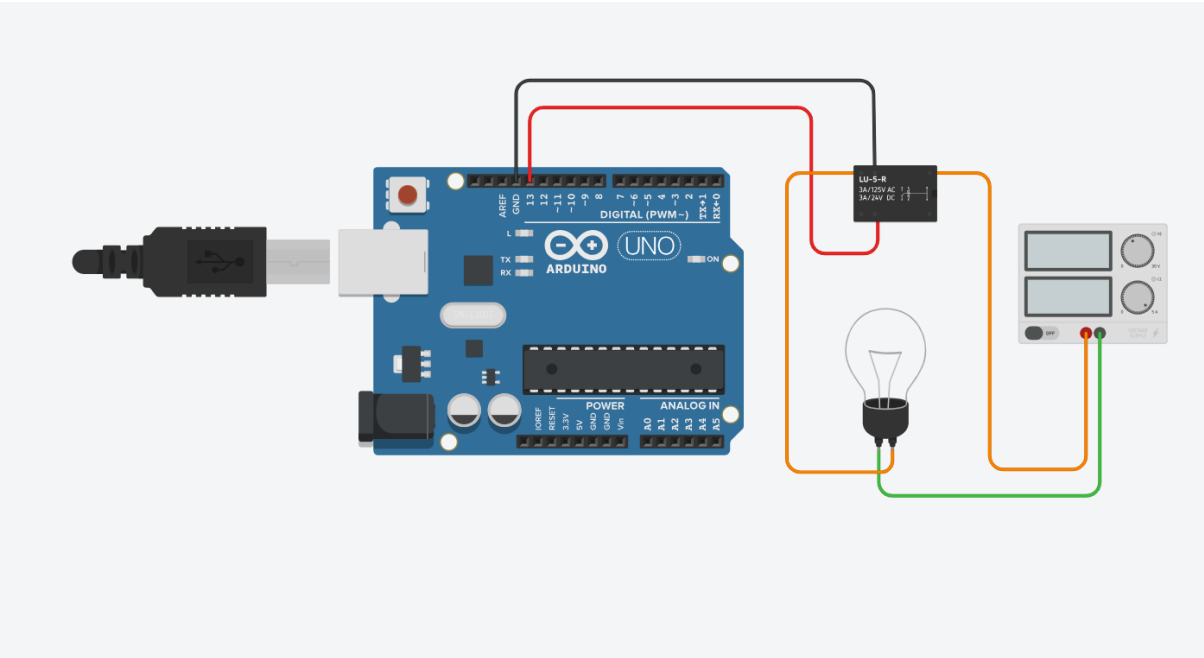
Aim

To interface a relay module with an Arduino board to control high voltage devices using Arduino's digital output pin.

Components:

Name	Quantity	Component
K1	1	Relay SPDT
U1	1	Arduino Uno R3
L1	1	Light bulb
P1	1	12 , 5 Power Supply

Circuits:



PROCEDURE:

Step 1: Open Tinkercad

1. Go to: https://www.tinkercad.com/
2. Log in or create a free account.
3. Click on **"Circuits"** from the left menu.
4. Click **"Create New Circuit"**.

Step 2: Add Components

Drag and drop the following components into your workspace:

- **Arduino Uno R3**
- **Relay Module** (5V)
- **Light Bulb**
- **DC Power Supply** (to simulate AC power)
- **Jumper wires**

Step 3: Wiring the Components

Follow this wiring diagram (as shown in the image):

➤ *Arduino to Relay:*

- **Digital Pin 7 → IN** pin of the Relay module
- **5V → VCC** on Relay
- **GND → GND** on Relay

➤ *Relay to Light Bulb & Power Supply:*

- **Power supply positive (+) → COM** on Relay
- **NO (Normally Open)** on Relay → **One terminal of the bulb**
- **Other terminal of the bulb → Power supply negative (-)**

Step 4: Add the Code

Click the **"Code"** button and use **Text (C++)** mode.

Program Code:

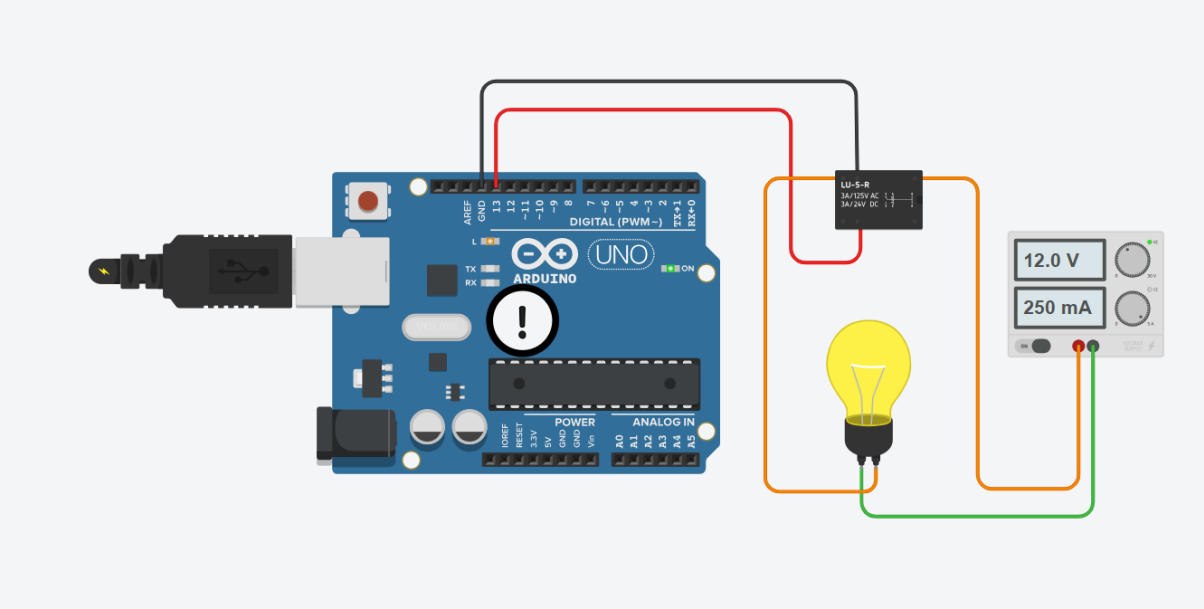
```
// C++ code
//
void setup()
{
  pinMode(13, OUTPUT);
}

void loop()
{
  digitalWrite(13, HIGH);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(13, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
}
```

Step 5: Start Simulation

1. Click **"Start Simulation"**.
2. The relay should click every 2 seconds, toggling the bulb ON and OFF.

Output:



Program 3:
Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

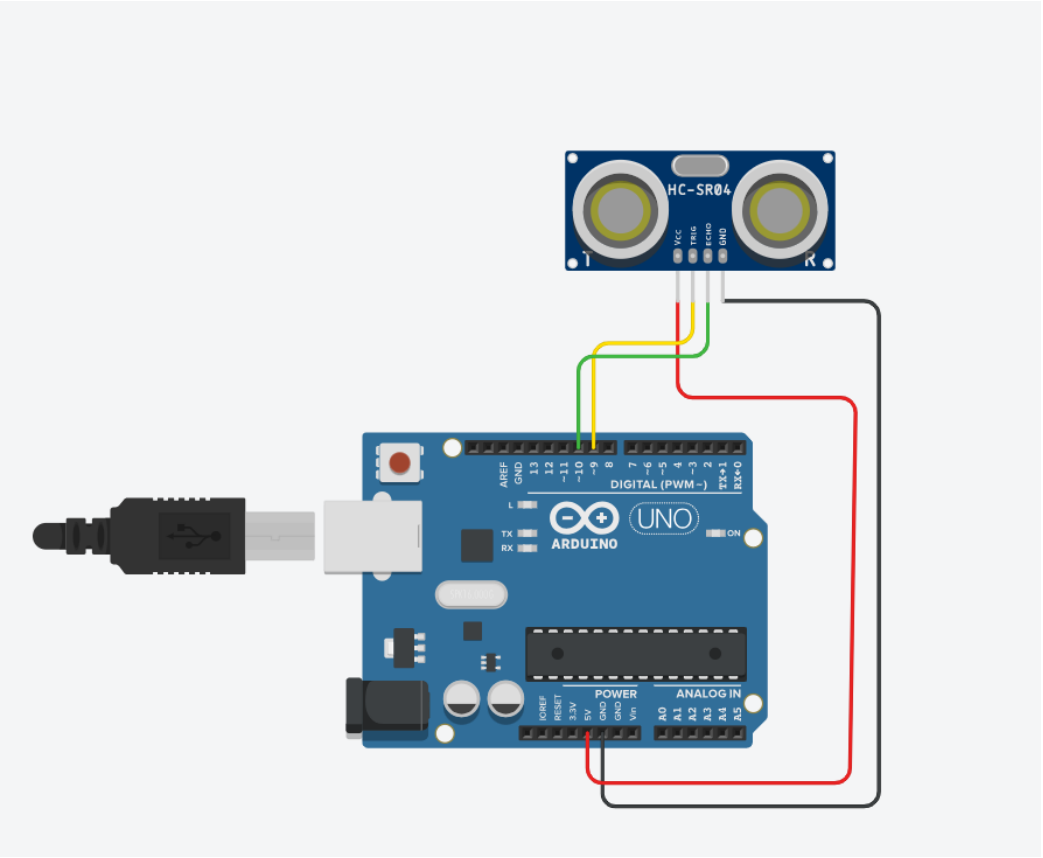
Aim

To develop an intrusion detection system using an ultrasonic sensor to detect movement within a certain distance and a sound sensor to detect noise, triggering an alarm or indicator when an intrusion is detected.

Components:

Name	Quantity	Component
U1	1	Arduino Uno R3
DIST1	1	Ultrasonic Distance Sensor (4-pin)

Circuit:



PROCEDURE:

1. Open Tinkercad:

- Go to https://www.tinkercad.com
- Log in or create a free account.

2. Create a New Circuit:

- Click on **"Circuits"** from the left-hand menu.
- Click **"Create New Circuit."**

3. Add Components:

Drag and drop the following components into your workspace:

- 1x **Arduino Uno R3**
- 1x **Ultrasonic Sensor HC-SR04**
- (Optional: 1x Breadboard if needed for future expansion)

4. Connect the Ultrasonic Sensor:

Referencing your image, do the wiring as follows:

HC-SR04 Pin	Connects To
VCC	Arduino **5V**
GND	Arduino **GND**
Trig	Arduino **Digital Pin 9**
Echo	Arduino **Digital Pin 10**

Use wires to make these connections. Match the colors as per your preference. The image uses:

- Red for VCC
- Black for GND
- Green for Echo
- Yellow for Trig

5. Add the Code:

Click the **"Code"** button (top-right), select **"Text" mode**, then paste the following Arduino code:

Program code:

```
// Pin definitions
#define trigPin 9
#define echoPin 10
#define soundSensor A0
#define alertPin 13  // LED or buzzer

void setup() {
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(alertPin, OUTPUT);
}

void loop() {
  // --- Ultrasonic Sensor Logic ---
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);

  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);

  long duration = pulseIn(echoPin, HIGH);
  int distance = duration * 0.034 / 2;

  // --- Sound Sensor Logic ---
  int soundLevel = analogRead(soundSensor);

  // --- Serial Output ---
  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.print(" cm | Sound: ");
  Serial.println(soundLevel);
```

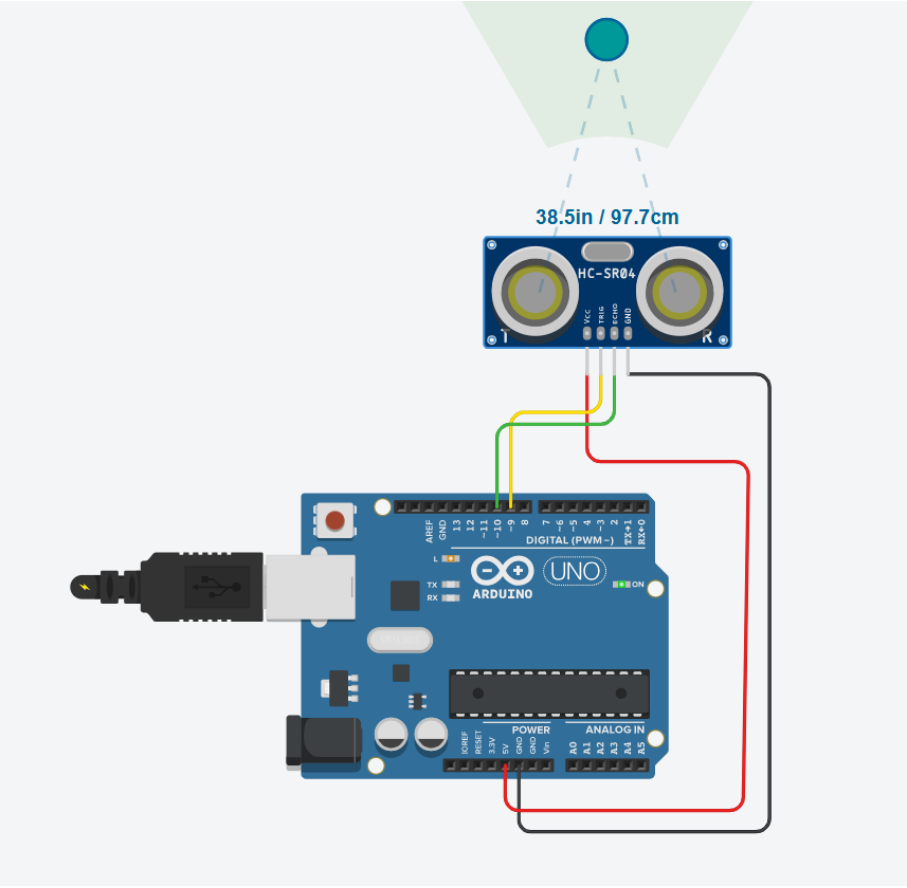
```
  // --- Alert Condition ---
  if ((distance > 0 && distance < 50) || soundLevel > 600) {
    digitalWrite(alertPin, HIGH); // Alert: Intruder detected
  } else {
    digitalWrite(alertPin, LOW);  // No intrusion
  }

  delay(300);
}
```

6. Start Simulation:

- Click **“Start Simulation.”**
- Open the **Serial Monitor** (click the small monitor icon or click the dropdown beside "Start Simulation").
- You should see the **distance readings in centimeters**.

Output:



Serial Monitor:
Distance: 96 cm | Sound: 387
Distance: 96 cm | Sound: 685
Distance: 96 cm | Sound: 101
Distance: 114 cm | Sound: 88
Distance: 125 cm | Sound: 414
Distance: 125 cm | Sound: 593
Distance: 125 cm | Sound: 928
Distance: 125 cm | Sound: 49
Distance: 125 cm | Sound: 318
Distance: 125 cm | Sound: 91
Distance: 125 cm | Sound: 86
Distance: 125 cm | Sound: 311
Distance: 125 cm | Sound: 687
Distance: 125 cm | Sound: 924
Distance: 125 cm | Sound: 793
Distance: 125 cm | Sound: 897
Distance: 125 cm | Sound: 206
Distance: 125 cm | Sound: 320
Distance: 125 cm | Sound: 583
Distance: 125 cm | Sound: 660
Distance: 125 cm | Sound: 561
Distance: 125 cm | Sound: 358
Distance: 125 cm | Sound: 979
Distance: 125 cm | Sound: 555
Distance: 125 cm | Sound: 184
Distance: 125 cm | Sound: 225
Distance: 125 cm | Sound: 516
Distance: 125 cm | Sound: 932
Distance: 125 cm | Sound: 989
Distance: 125 cm | Sound: 1006
Distance: 125 cm | Sound: 717
Distance: 125 cm | Sound: 768

Program 4:

Develop a program to control a DC motor with Arduino board.

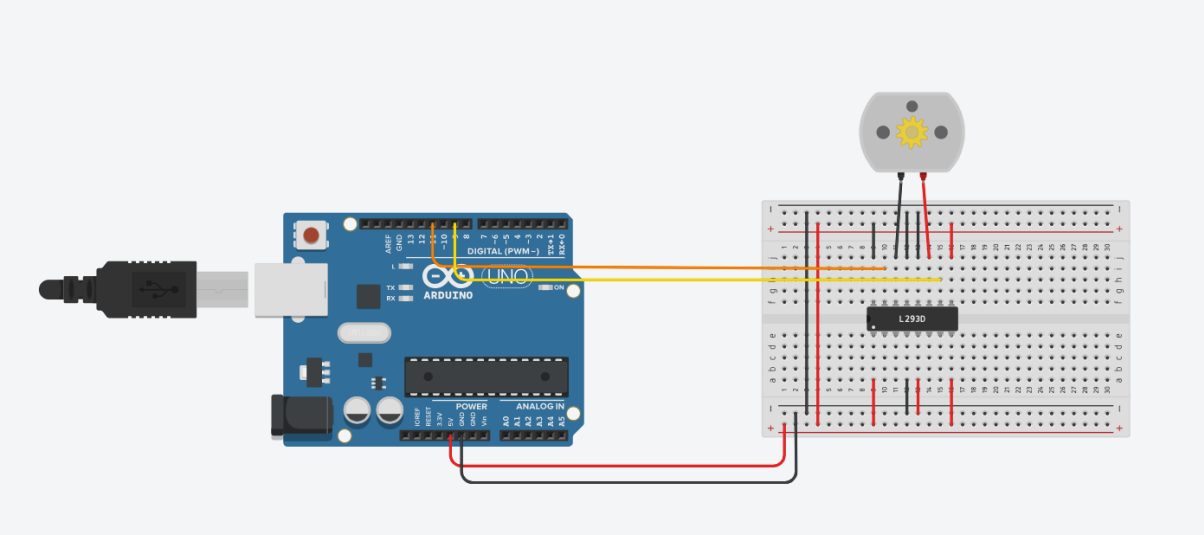
AIM

To develop a program that controls a DC motor using an Arduino board.

Component:

Name	Quantity	Component
U1	1	Arduino Uno R3
M1	1	DC Motor
U2	1	H-bridge Motor Driver

Circuit :



PROCEDURE:

STEP 1: Open Tinkercad

- Go to: https://www.tinkercad.com
- Log in → Click **"Circuits"** → Click **"Create New Circuit"**

STEP 2: Add Components

From the components menu, drag and drop:

- **Arduino Uno R3**
- **Breadboard**
- **DC Motor**
- **L298N Motor Driver (IC)**
- Jumper Wires

STEP 3: Make the Connections

🔌 L298N Motor Driver IC Pin Connections:

L298N Pin	Connects To
Pin 1 (Enable A)	Arduino Digital Pin **9**
Pin 2 (Input 1)	Arduino Digital Pin **8**
Pin 3 (Output 1)	One terminal of **DC Motor**
Pin 4 (GND)	**GND** (to Breadboard/Arduino)
Pin 5 (GND)	**GND** (to Breadboard/Arduino)
Pin 6 (Output 2)	Other terminal of **DC Motor**
Pin 7 (Input 2)	Arduino Digital Pin **7**
Pin 8 (VCC for Motor)	External **5V (or 9V battery)** (optional, or use 5V from Arduino if low-power motor)
Pin 9 (VCC for IC)	**5V** from Arduino

Make sure both **GNDs** of the Arduino and L298N are connected together (common ground).

STEP 4: Upload the Code

Click **"Code"** → **Text Mode** → Paste this Arduino code:

Program Code:

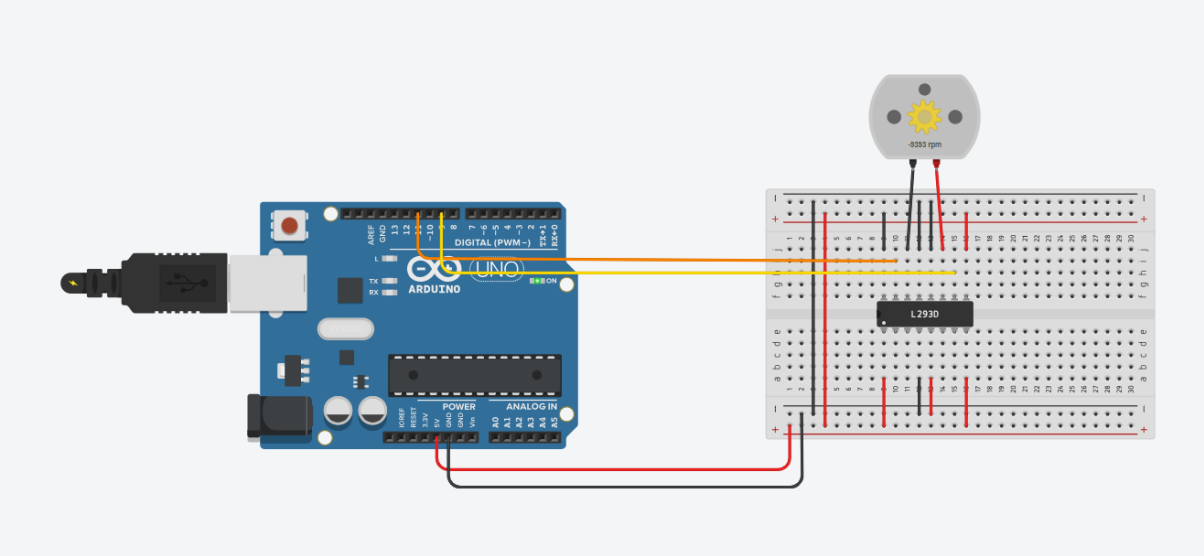
```
// C++ code
//
void setup()
{
  pinMode(11, OUTPUT);
  pinMode(9,OUTPUT);
}

void loop()
{
  digitalWrite(11, HIGH);
  digitalWrite(9,LOW);
  delay(3000); // Wait for 1000 millisecond(s)
  digitalWrite(11, LOW);
  digitalWrite(9,LOW);
  delay(3000);
  digitalWrite(9,HIGH);
  digitalWrite(11, LOW);
  delay(3000); // Wait for 1000 millisecond(s)
  digitalWrite(11, HIGH);
  digitalWrite(9,LOW);
  delay(3000);
}
```

STEP 5: Start the Simulation

- Click **"Start Simulation"**
- The **motor should spin forward for 2 seconds**, pause, **reverse for 2 seconds**, then pause again in a loop

Output:



Program 5:
Develop a program to deploy smart street light system using LDR sensor.

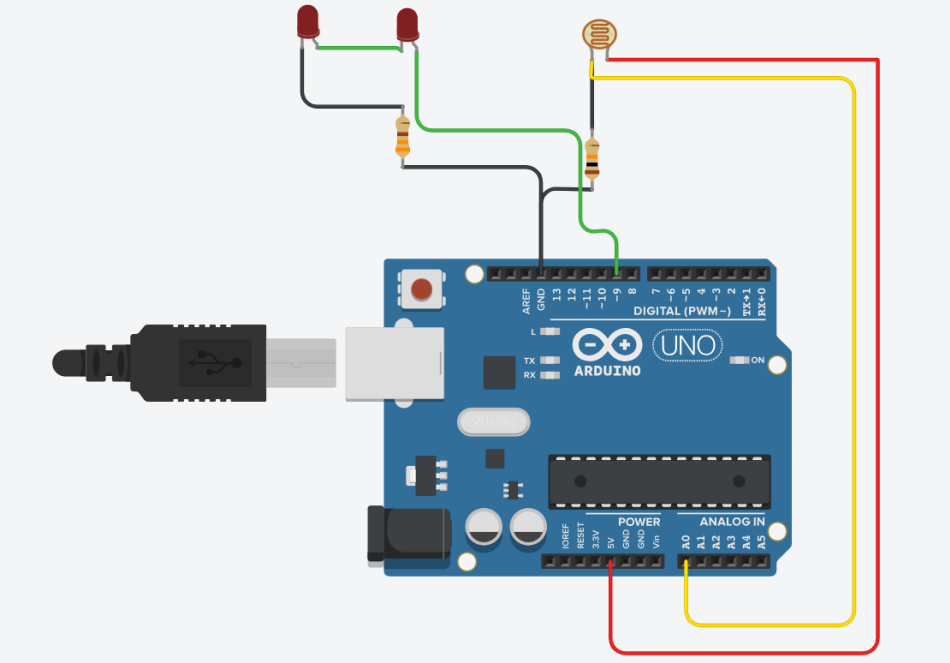
AIM

To develop a program to deploy a smart street light system using an LDR (Light Dependent Resistor) sensor with an Arduino board.

Components:

Name	Quantity	Component
U1	1	Arduino Uno R3
R1	1	Photoresistor
D1 D3	2	Red LED
R2	1	330 Ω Resistor
R3	1	10 kΩ Resistor

Circuit:



PROCEDURE:

STEP 1: Open Tinkercad

1. Go to https://www.tinkercad.com
2. Log in → Click **"Circuits"** → Click **"Create New Circuit"**

STEP 2: Add Components

Drag and drop the following into the workspace:

- **Arduino Uno R3**
- **LDR sensor**
- **2 LEDs**
- **Resistors (220Ω and 10kΩ)**
- Jumper wires

STEP 3: Make the Circuit Connections

📷 *LDR Circuit (Light Sensor)*

- Connect **one leg of the LDR** to **5V**
- Connect **the other leg of the LDR** to:
  - **A0 (Analog pin)** of Arduino
  - **One side of 10kΩ resistor**
- Connect the **other side of the 10kΩ resistor** to **GND**

This forms a **voltage divider** for reading light levels on A0.

💡 *LEDs Circuit*

- Connect **anode (long leg)** of each LED to **Digital Pin 8 and 9**
- Connect **cathode (short leg)** of each LED to **GND through a 220Ω resistor**

LED	Anode →	Cathode →
LED1	Pin 8	GND via 220Ω

LED	Anode →	Cathode →
LED2	Pin 9	GND via 220Ω

STEP 4: Add the Arduino Code

Click on **"Code" → Select "Text" mode** → Paste this code:

Program code:

```
const int LDR_PIN = A0;
const int LED_PIN = 9;
int threshold = 600;  // Adjust after Serial Monitor test

void setup() {
  pinMode(LED_PIN, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int ldrValue = analogRead(LDR_PIN);
  Serial.println(ldrValue);  // See values in Serial Monitor

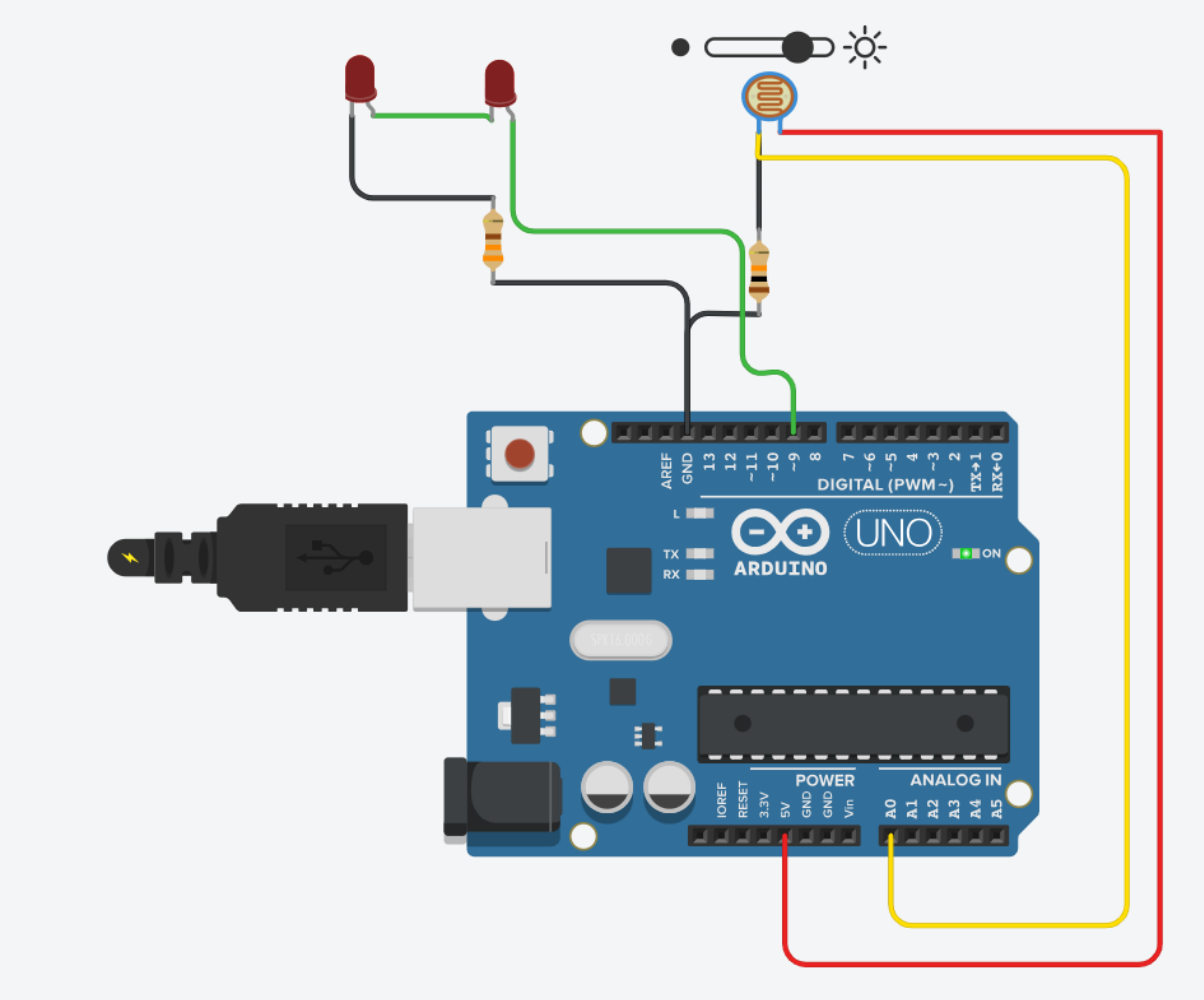
  if (ldrValue < threshold) {
    digitalWrite(LED_PIN, HIGH);  // Turn ON in dark
  } else {
    digitalWrite(LED_PIN, LOW);   // Turn OFF in light
  }

  delay(500);
}
```

STEP 5: Start the Simulation

- Click **"Start Simulation"**
- Open the **Serial Monitor** to view the LDR value
- **Block the LDR** (with your cursor or simulate dark conditions)
  - LED1 should turn ON (LED2 OFF) in dark
  - LED2 should turn ON (LED1 OFF) in bright light

Output:


AREF
GND
13
12
~11
~10
~9
8
7
~6
~5
4
~3
2
TX→1
RX←0
DIGITAL (PWM~)
L
TX
RX
ARDUINO
UNO
ON
POWER
ANALOG IN
IOREF
RESET
3.3V
5V
GND
GND
Vin
A0
A1
A2
A3
A4
A5

Program 6:

Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).

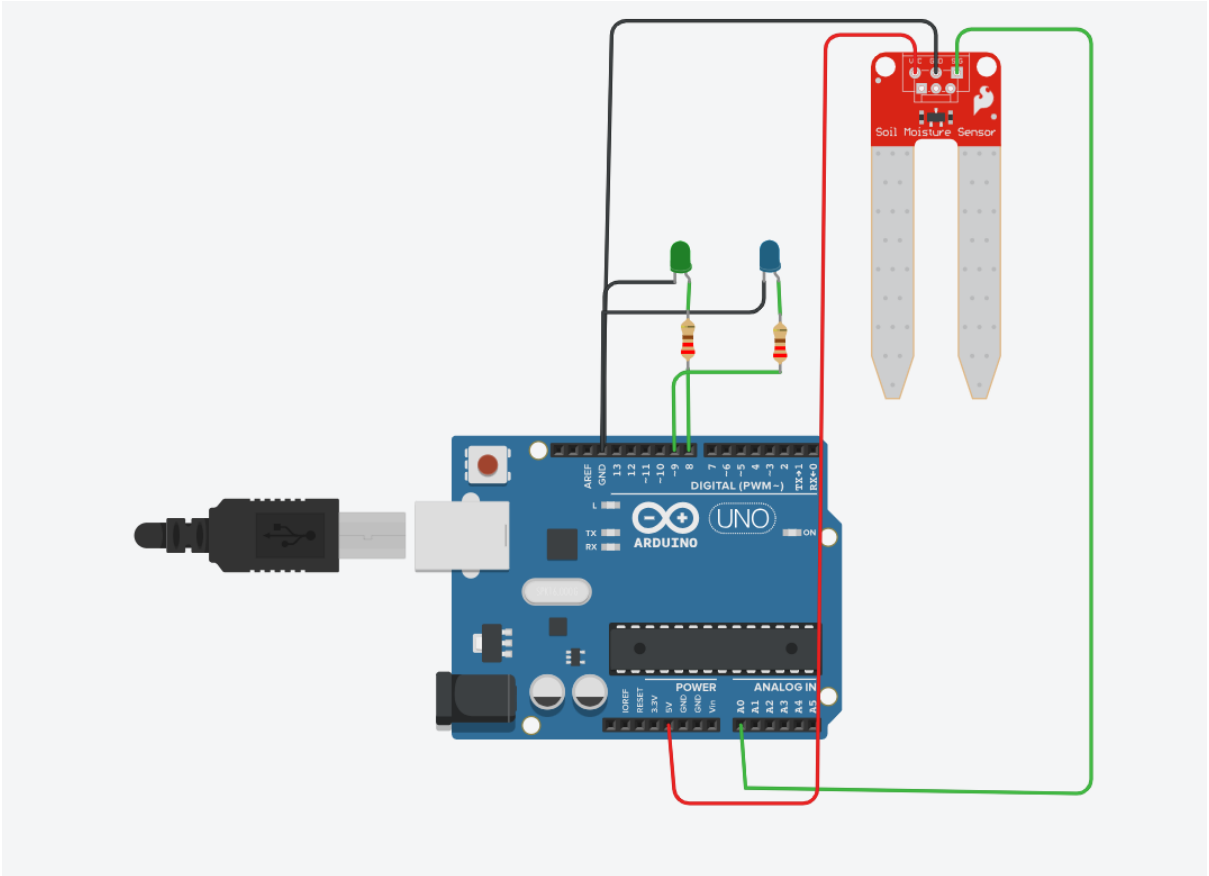
AIM:

Develop a program to classify dry and wet waste using a DHT22 sensor based on humidity readings.

Components:

Name	Quantity	Component
U1	1	Arduino Uno R3
SEN1	1	Soil Moisture Sensor
D1	1	Green LED
D2	1	Blue LED
R1 R2	2	220 Ω Resistor

Circuit:



PROCEDURE:

STEP 1: Open Tinkercad

1. Go to https://www.tinkercad.com
2. Login → Go to **"Circuits"** → Click **"Create New Circuit"**

STEP 2: Add Components

Drag the following components into your circuit:

- **Arduino Uno R3**
- **Soil Moisture Sensor (with fork and amplifier module)**
- **2 LEDs (Green and Blue)**
- **2 Resistors (220Ω)**
- Jumper wires

STEP 3: Wire the Circuit

Soil Moisture Sensor Connections:

Soil Sensor Pin	Connects To
VCC	Arduino **5V**
GND	Arduino **GND**
A0 (Analog Out)	Arduino **A0**

LED Connections:

LED Color	Anode (Long Leg) →	Cathode →
Green	Digital Pin **8**	GND via 220Ω
Blue	Digital Pin **9**	GND via 220Ω

Use resistors in series with the LEDs to prevent burning them out.

STEP 4: Add the Arduino Code

Click **“Code” → Select “Text” mode** and paste the following code:

Program code:

```
const int MOISTURE_PIN = A0;  // Sensor analog pin
const int DRY_LED = 8;        // Green LED for dry waste
const int WET_LED = 9;        // Blue LED for wet waste

int threshold = 600; // Tune this value based on readings

void setup() {
  Serial.begin(9600);
  pinMode(DRY_LED, OUTPUT);
  pinMode(WET_LED, OUTPUT);
}

void loop() {
  int moistureValue = analogRead(MOISTURE_PIN);
  Serial.print("Moisture: ");
  Serial.println(moistureValue);

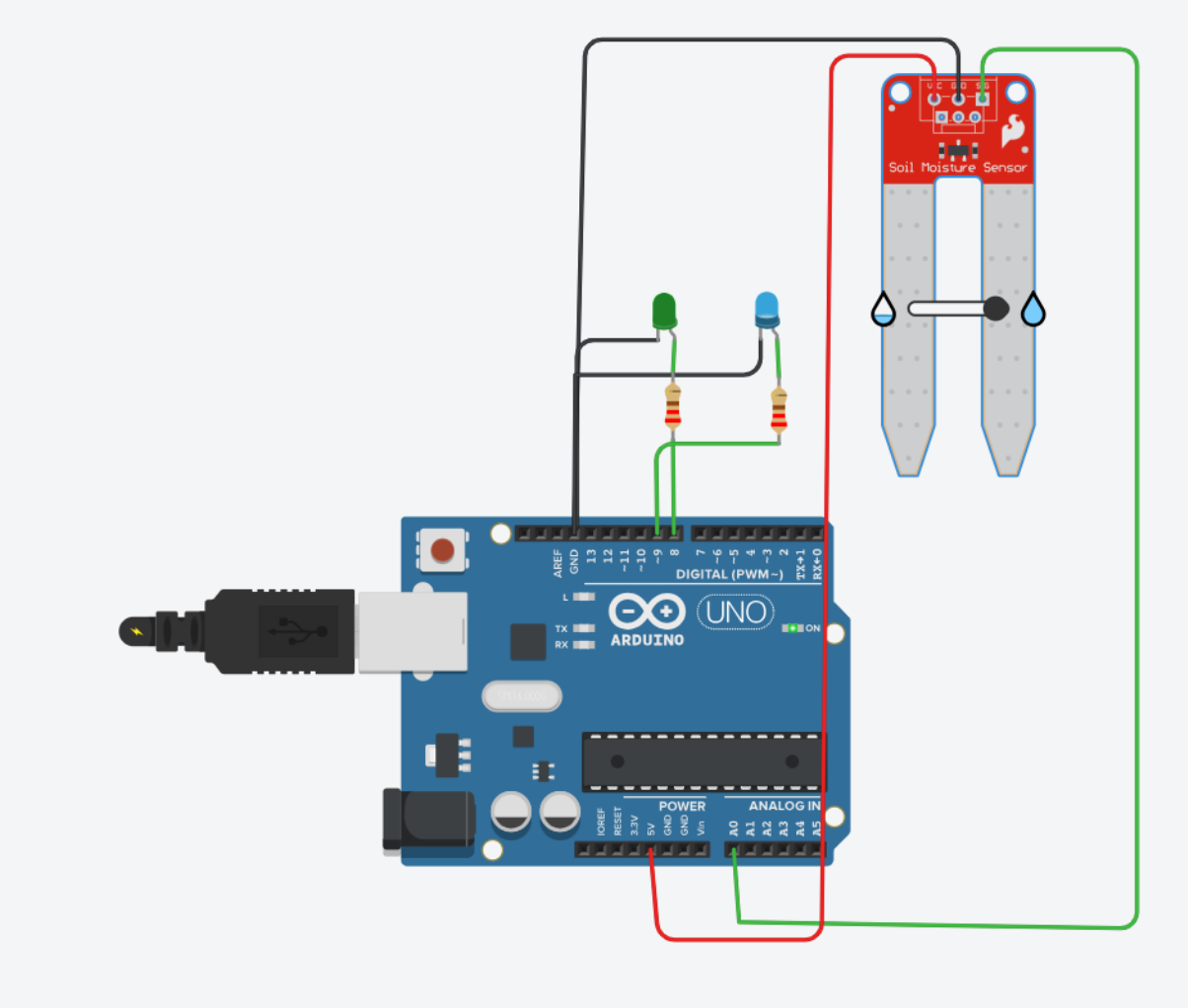
  if (moistureValue > threshold) {
    // Wet waste
    Serial.println("Classified: WET WASTE");
    digitalWrite(WET_LED, HIGH);
    digitalWrite(DRY_LED, LOW);
  } else {
    // Dry waste
    Serial.println("Classified: DRY WASTE");
    digitalWrite(DRY_LED, HIGH);
    digitalWrite(WET_LED, LOW);
  }

  delay(1000);  // Wait for 1 second
}
```

STEP 5: Start the Simulation

- Click **“Start Simulation”**
- Open the **Serial Monitor** to view real-time moisture values
- You can **simulate wet/dry conditions** by changing the soil sensor’s value in the simulation settings

OUTPUT:



Program 7:
Develop a program to read the pH value of a various substances like milk, lime and water.

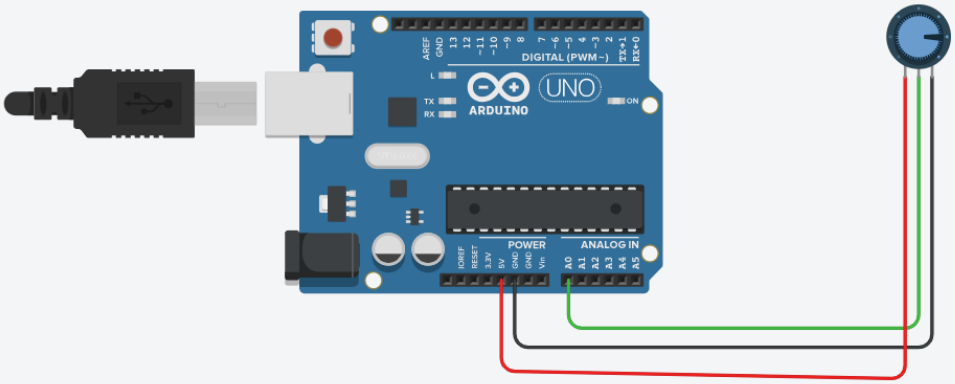
AIM

To develop a program that reads the pH value of various substances such as milk, lime juice, and water using a pH sensor and Arduino board.

Components:

Name	Quantity	Component
U1	1	Arduino Uno R3
Rpot1	1	250 kΩ Potentiometer

Circuit:



PROCEDURE:

STEP 1: Open Tinkercad

1. Visit https://www.tinkercad.com
2. Log in → Go to **"Circuits"** → Click **"Create New Circuit"**

STEP 2: Add Components

Drag the following components into your workspace:

- **Arduino Uno R3**
- **Potentiometer**
- **Jumper wires**

STEP 3: Wiring the Potentiometer

A potentiometer has **3 pins**:

- **Left pin** → Connect to **GND**
- **Right pin** → Connect to **5V**
- **Middle pin (wiper)** → Connect to **Analog Pin A0**

Wiring summary based on your image:

Potentiometer Pin	Connects To
Left	GND (Black wire)
Right	5V (Red wire)
Middle (wiper)	A0 (Green wire)

STEP 4: Add the Arduino Code

Click **"Code" → Select "Text" Mode** → Paste the following code:

Program Code:

```
const int PH_PIN = A0;  // Simulated pH analog input

void setup() {
  Serial.begin(9600);
}

void loop() {
  int analogValue = analogRead(PH_PIN);      // Range: 0–1023
  float voltage = analogValue * (5.0 / 1023.0); // Convert to voltage (0–5V)

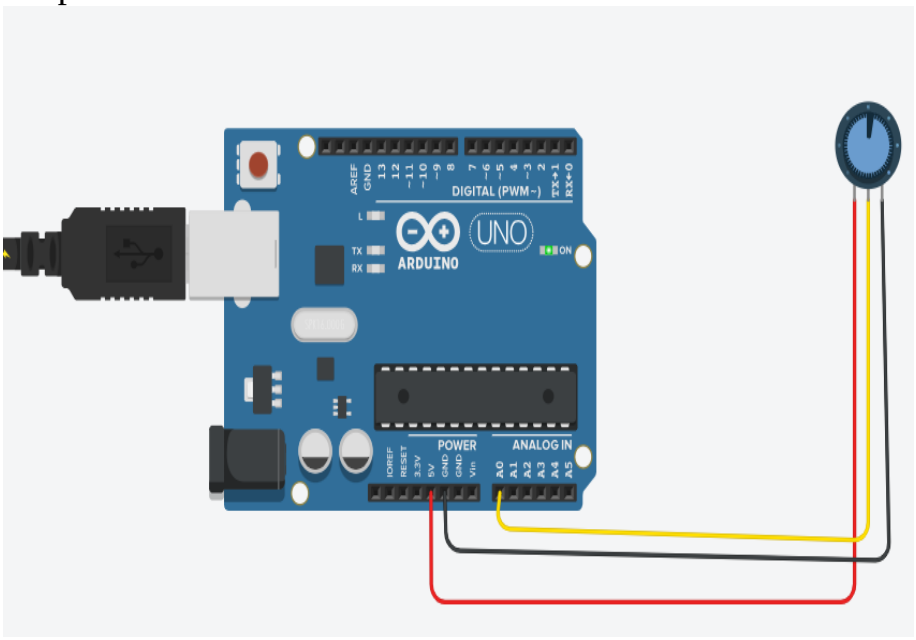
  // Simulate pH scale: 0V = pH 0, 5V = pH 14
  float pHValue = voltage * (14.0 / 5.0);

  Serial.print("Analog: ");
  Serial.print(analogValue);
  Serial.print(" | Voltage: ");
  Serial.print(voltage);
  Serial.print(" V | pH Value: ");
  Serial.println(pHValue);

  // Classify substance based on pH
  if (pHValue < 3.0) {
    Serial.println("Substance: Lime Juice (Strong Acid)");
  } else if (pHValue >= 6.0 && pHValue < 7.0) {
    Serial.println("Substance: Milk (Weak Acid)");
  } else if (pHValue >= 7.0 && pHValue <= 7.5) {
    Serial.println("Substance: Water (Neutral)");
  } else {
    Serial.println("Substance: Unknown");
  }
```

```
  delay(1000);
}
```

STEP 5: Start the Simulation

- Click **“Start Simulation”**
- Open the **Serial Monitor** (click the monitor icon)
- **Turn the potentiometer knob** in the simulation
    - You will see values ranging from **0 (min)** to **1023 (max)** based on the position
    - 

Output:
Substance: Unknown
Analog: 818 | Voltage: 4.00 V | pH Value: 11.19
Substance: Unknown
Analog: 818 | Voltage: 4.00 V | pH Value: 11.19
Substance: Unknown
Analog: 818 | Voltage: 4.00 V | pH Value: 11.19
Substance: Unknown
Analog: 143 | Voltage: 0.70 V | pH Value: 1.96
Substance: Lime Juice (Strong Acid)
Analog: 143 | Voltage: 0.70 V | pH Value: 1.96
Substance: Lime Juice (Strong Acid)
Analog: 164 | Voltage: 0.80 V | pH Value: 2.24

Substance: Lime Juice (Strong Acid)
Analog: 164 | Voltage: 0.80 V | pH Value: 2.24
Substance: Lime Juice (Strong Acid)
Analog: 164 | Voltage: 0.80 V | pH Value: 2.24
Substance: Lime Juice (Strong Acid)
Analog: 164 | Voltage: 0.80 V | pH Value: 2.24
Substance: Lime Juice (Strong Acid)
Analog: 164 | Voltage: 0.80 V | pH Value: 2.24
Substance: Lime Juice (Strong Acid)
Analog: 164 | Voltage: 0.80 V | pH Value: 2.24
Substance: Lime Juice (Strong Acid)

Program 8:

Develop a program to detect the gas leakage in the surrounding environment.

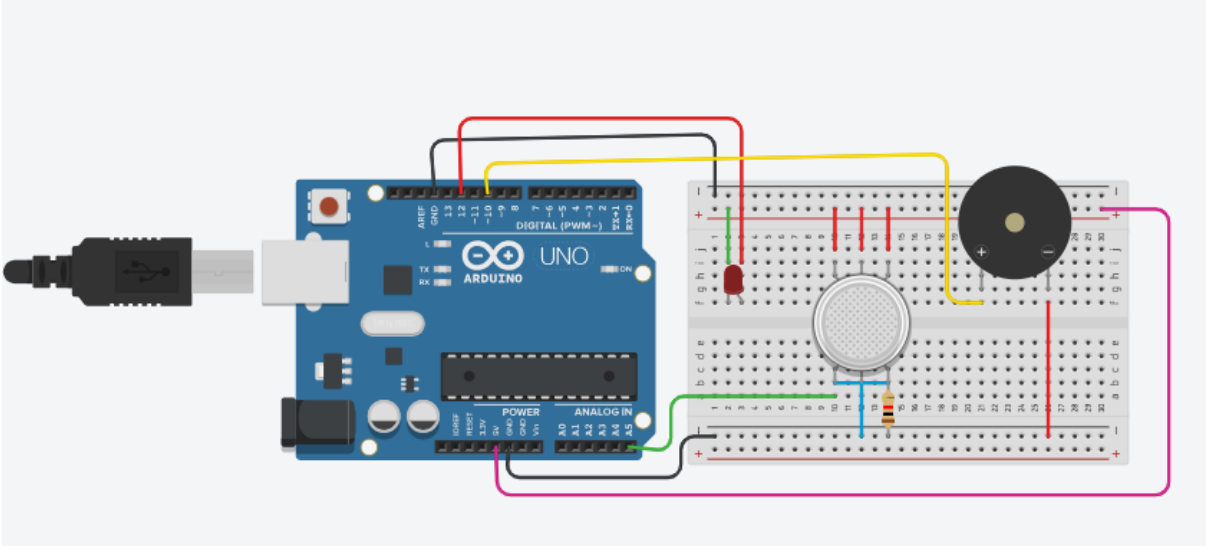
AIM

To develop a program to detect gas leakage in the surrounding environment using a gas sensor and Arduino.

Component:

Name	Quantity	Component
U1	1	Arduino Uno R3
PIEZO1	1	Piezo
D3	1	Red LED
GAS1	1	Gas Sensor
R1	1	1 kΩ Resistor

Circuit:



PROCEDURE:

STEP 1: Open Tinkercad

1. Go to https://www.tinkercad.com
2. Log in → Go to **"Circuits"** → Click **"Create New Circuit"**

STEP 2: Add Components

From the components panel, add:

- **Arduino Uno R3**
- **MQ-2 Gas Sensor**
- **Red LED**
- **Buzzer**
- **Breadboard**
- **Resistors (220Ω, 10kΩ)**
- **Jumper wires**

STEP 3: Wiring the Components

MQ-2 Gas Sensor Connections (Analog Output)

MQ-2 Pin	Connects To
VCC	Arduino **5V**

MQ-2 Pin	Connects To
GND	Arduino **GND**
A0	Arduino **A0**

Some versions may also have a **DO (digital out)** — use **A0** for analog input.

LED Connection

LED Leg	Connects To
Anode (+)	Arduino **Pin 8** (via **220Ω resistor**)
Cathode (–)	GND

Buzzer Connection

Buzzer Pin	Connects To
+ (long pin)	Arduino **Pin 9**
– (short pin)	GND

STEP 4: Add the Arduino Code

Click **"Code" → Select "Text" mode →** Paste this code:

Program code:

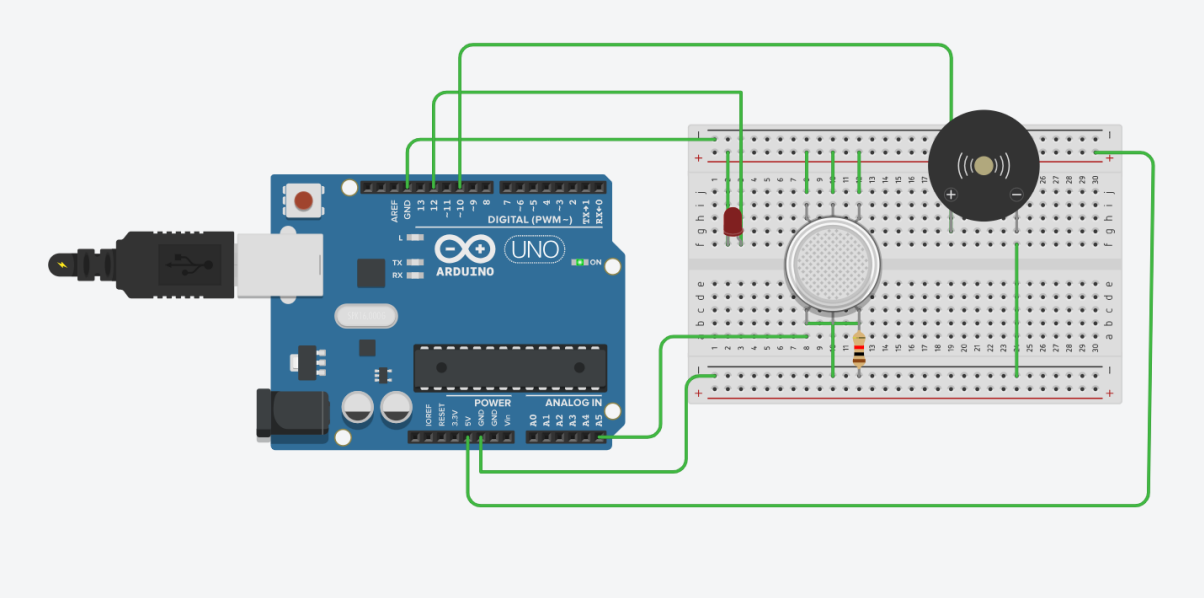
```
int redLed=12;
int buzzer=10;
int smokeA0=A5;
int sensorThree=100;
void setup()
{
  pinMode(redLed,OUTPUT);
  pinMode(buzzer,OUTPUT);
  pinMode(smokeA0,INPUT);
  Serial.begin(9600);
}
void  loop()
```

```
{
 int analogSensor=analogRead(smokeA0);
 Serial.print("Pin A0:");
 Serial.println(analogSensor);
 if(analogSensor>sensorThree)
 {
  digitalWrite(redLed,HIGH);
  tone(buzzer,1000,200);
 }
 else
 {
  digitalWrite(redLed,LOW);
  noTone(buzzer);
 }
 delay(100);
}
```

STEP 5: Start the Simulation

- Click **“Start Simulation”**
- Open the **Serial Monitor**
- Increase the analog value of the **MQ sensor** in the simulation settings to simulate gas detection
    - Once it goes above **400**, the **LED and buzzer turn ON**

Output:



Pin A0:122
Pin A0:122
Pin A0:122
Pin A0:130
Pin A0:146
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
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Pin A0:85
Pin A0:85
Pin A0:90
Pin A0:121
Pin A0:154
Pin A0:150
Pin A0:149
Pin A0:149
Pin A0:181
Pin A0:251
Pin A0:243
Pin A0:277
Pin A0:224
Pin A0:136

Pin A0:124
Pin A0:119
Pin A0:114
Pin A0:123
Pin A0:142
Pin A0:130
Pin A0:100
Pin A0:90
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:85
Pin A0:87
Pin A0:133
Pin A0:142
Pin A0:142
Pin A0:147
Pin A0:149
Pin A0:150

Program 9:
Develop a program to demonstrate weather station readings using Arduino.

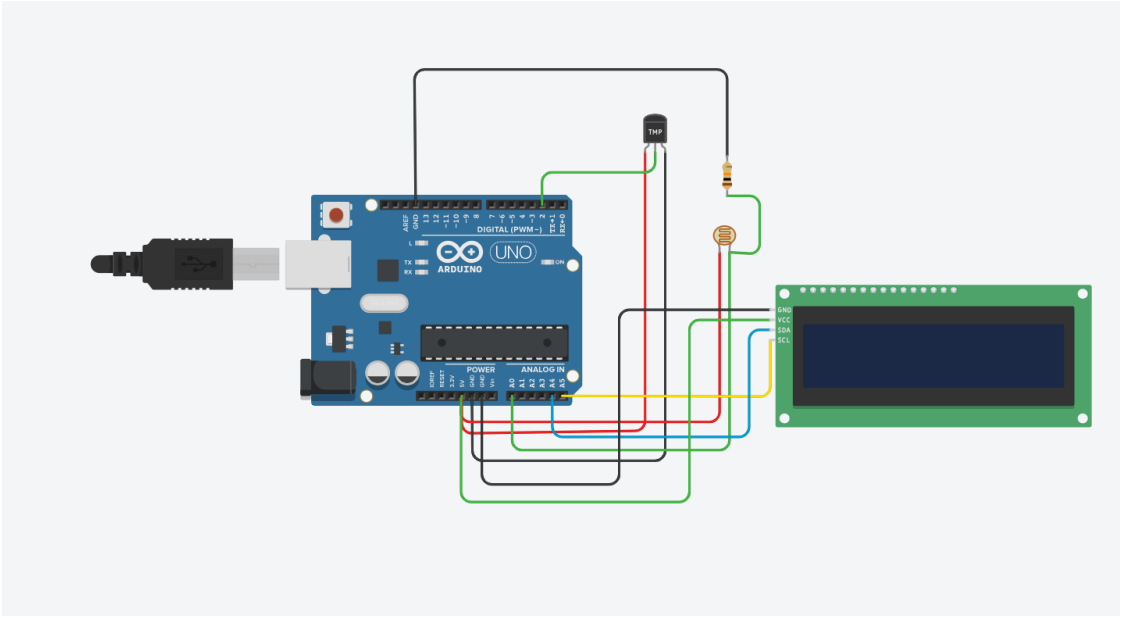
AIM

To develop a program to demonstrate a weather station using Arduino to measure temperature, humidity, and optionally atmospheric pressure.

Components:

Name	Quantity	Component
U1	1	Arduino Uno R3
U2	1	Temperature Sensor [TMP36]
R1	1	Photoresistor
R2	1	10 kΩ Resistor
U5	1	MCP23008-based, 32 (0x20) LCD 16 x 2 (I2C)

Circuit:



PROCEDURE:

STEP 1: Open Tinkercad

1. Go to https://www.tinkercad.com
2. Log in → Go to **"Circuits"** → Click **"Create New Circuit"**

STEP 2: Add Components

Search and drag the following into your workspace:

- Arduino Uno R3
- **LM35 Temperature Sensor**
- **16x2 LCD Display (standard)**
- **10kΩ potentiometer**
- Breadboard
- Jumper wires

STEP 3: Wire the Circuit

LM35 Temperature Sensor Wiring

LM35 Pin	Connects To
Left (VCC)	Arduino **5V**
Middle (Vout)	Arduino **A0**
Right (GND)	Arduino **GND**

LCD to Arduino Wiring (in 4-bit mode)

LCD Pin	Connects To
VSS	GND
VDD	5V

LCD Pin		Connects To
VO		Middle pin of potentiometer (for contrast)
RS		Pin **12**
RW		GND
E		Pin **11**
D4		Pin **5**
D5		Pin **4**
D6		Pin **3**
D7		Pin **2**
A	(LED+)	5V (optional for backlight)
K	(LED-)	GND

🔁 *Potentiometer Connections*

- One side → **GND**
- Other side → **5V**
- Middle → LCD **VO** pin (for contrast)

STEP 4: Add the Arduino Code

Click **“Code” → Select “Text” mode** → Paste the following code:

Program code:

```
#include <LiquidCrystal.h>

// LCD pin setup: RS, E, D4, D5, D6, D7
LiquidCrystal lcd(7, 6, 5, 4, 3, 8);

// TMP36 on A1, LDR on A0
const int tempPin = A1;
const int ldrPin = A0;

void setup() {
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.print("Weather Station");
  delay(2000);
  lcd.clear();
}

void loop() {
  // Read temperature from TMP36
  int tempReading = analogRead(tempPin);
  float voltage = tempReading * (5.0 / 1023.0);
  float temperatureC = (voltage - 0.5) * 100;

  // Read LDR and map to light percentage
  int ldrValue = analogRead(ldrPin);
  int lightPercent = map(ldrValue, 0, 1023, 100, 0); // 0 = bright, 100 = dark

  // Print to Serial Monitor
  Serial.print("Temp: ");
  Serial.print(temperatureC);
  Serial.print(" °C, Light: ");
  Serial.print(lightPercent);
  Serial.println("%");

  // Display on LCD
```

```
  lcd.setCursor(0, 0);
  lcd.print("Temp:");
  lcd.print(temperatureC, 1);
  lcd.print((char)223); // Degree symbol
  lcd.print("C     ");

  lcd.setCursor(0, 1);
  lcd.print("Light:");
  lcd.print(lightPercent);
  lcd.print("%     ");

  delay(2000);
}
```

STEP 5: Start the Simulation

- Click **“Start Simulation”**
- You should see **temperature values displayed on the LCD screen**
- You can simulate different temperatures by adjusting the LM35 in the Tinkercad panel

Output:
Temp: -33.87 Â°C, Light: 95%
Temp: 291.15 Â°C, Light: 95%
Temp: 262.81 Â°C, Light: 95%
Temp: 232.50 Â°C, Light: 95%
Temp: 310.70 Â°C, Light: 95%
Temp: 209.53 Â°C, Light: 95%

Program 10:
Develop a program to setup a UART protocol and pass a string through the protocol.

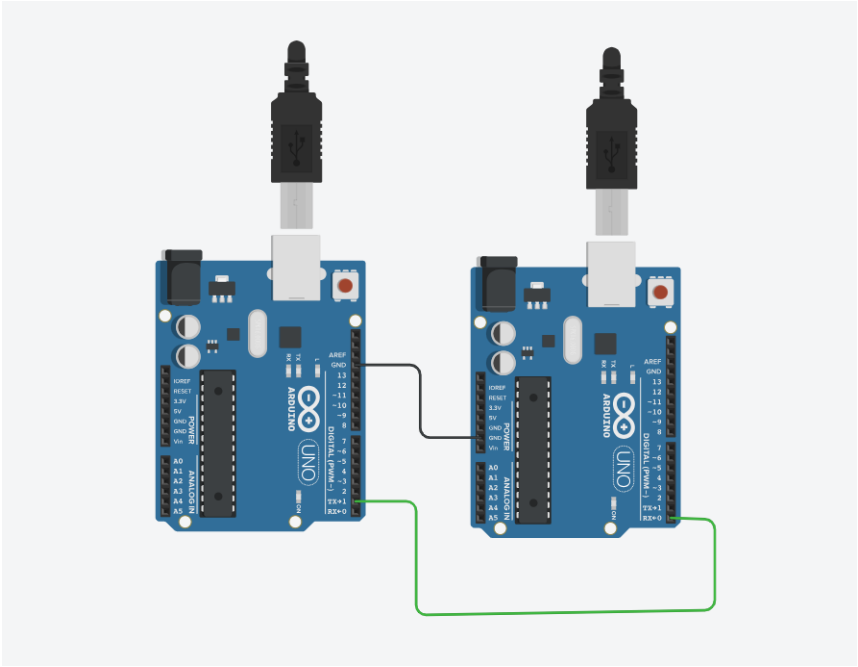
AIM

To develop a program that sets up UART communication and transmits a string using the UART protocol.

Components:

Name	Quantity	Component
U1 U2	2	Arduino Uno R3

Circuit:



PROCEDURE:

STEP 1: Open Tinkercad

1. Go to https://www.tinkercad.com
2. Log in → Go to **Circuits** → Click **Create New Circuit**

STEP 2: Add Components

- Drag **2 Arduino Uno** boards into the workspace.

STEP 3: Connect the Arduinos

Connection	Arduino 1 Pin	Arduino 2 Pin
Ground (GND)	GND	GND
TX → RX	Pin 1 (TX)	Pin 0 (RX)
RX → TX	Pin 0 (RX)	Pin 1 (TX)

In your image, only one data wire and GND wire are connected; typically, TX of Arduino1 goes to RX of Arduino2 and vice versa for full duplex.

STEP 4: Upload Arduino Sketches

Program code:

Arduino 1 (Sender):

```
void setup() {
  Serial.begin(9600);
}

void loop() {
  Serial.println("Hello from Arduino 1");
  delay(1000);
}
```

Arduino 2 (Receiver):

Code for Receiver (Arduino 2):

```
void setup() {
  Serial.begin(9600);
}

void loop() {
  if (Serial.available()) {
    String msg = Serial.readStringUntil('\n');
    Serial.print("Received: ");
    Serial.println(msg);
  }
}
```

STEP 5: Simulate

- Upload sketches to respective Arduinos (one sender, one receiver).
- Start simulation.
- Open **Serial Monitor** for Arduino 2 to see incoming messages.

Output:
Sender (Arduino 1):
Hello from Arduino 1
Hello from Arduino 1
Receiver (Arduino 2):
Received: Hello from Arduino 1
Received: Hello from Arduino 1

Program 11:
Develop a water level depth detection system using Ultrasonic sensor.

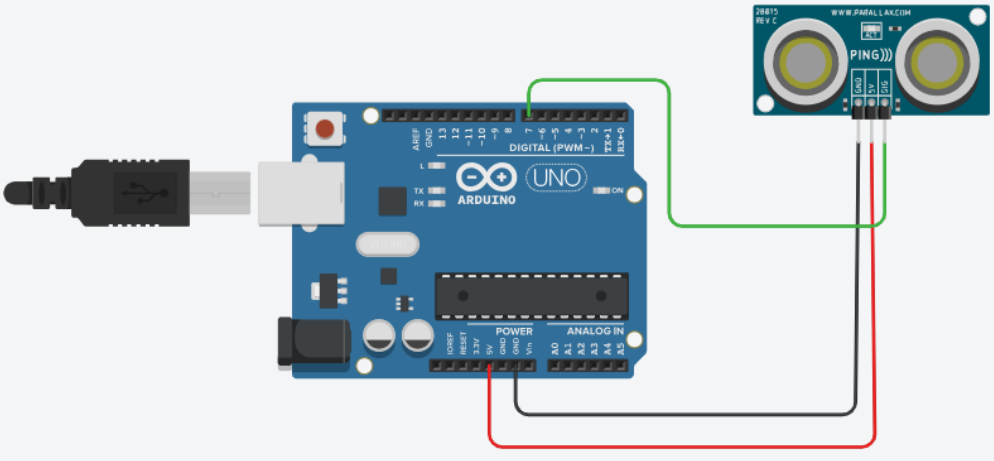
AIM

To develop a system to detect water level depth using an ultrasonic sensor and Arduino.

Components:

Name	Quantity	Component
U1	1	Arduino Uno R3
PING1	1	Ultrasonic Distance Sensor

Circuit:



PROCEDURE:

STEP 1: Open Tinkercad

1. Go to https://www.tinkercad.com
2. Log in → Go to **Circuits** → Click **Create New Circuit**

STEP 2: Add Components

- Drag **Arduino Uno** and **HC-SR04 Ultrasonic Sensor** into the workspace.

STEP 3: Wiring the HC-SR04

HC-SR04 Pin	Connects To Arduino Pin
VCC	5V
GND	GND
Trig	Digital Pin 7
Echo	Digital Pin 6

STEP 4: Add Arduino Code

Click **"Code" → Select "Text" mode** → Paste this code:

Program Code:

```
const int sensorPin = 7; // SIG pin

long duration;
float distanceCM;

void setup() {
  Serial.begin(9600);
  pinMode(sensorPin, OUTPUT);
}

void loop() {
```

```
  // Send trigger pulse
  pinMode(sensorPin, OUTPUT);
  digitalWrite(sensorPin, LOW);
  delayMicroseconds(2);
  digitalWrite(sensorPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(sensorPin, LOW);

  // Change pin to input to read echo
  pinMode(sensorPin, INPUT);

  // Read the echo pulse duration
  duration = pulseIn(sensorPin, HIGH);

  // Calculate distance in cm
  distanceCM = duration * 0.034 / 2;

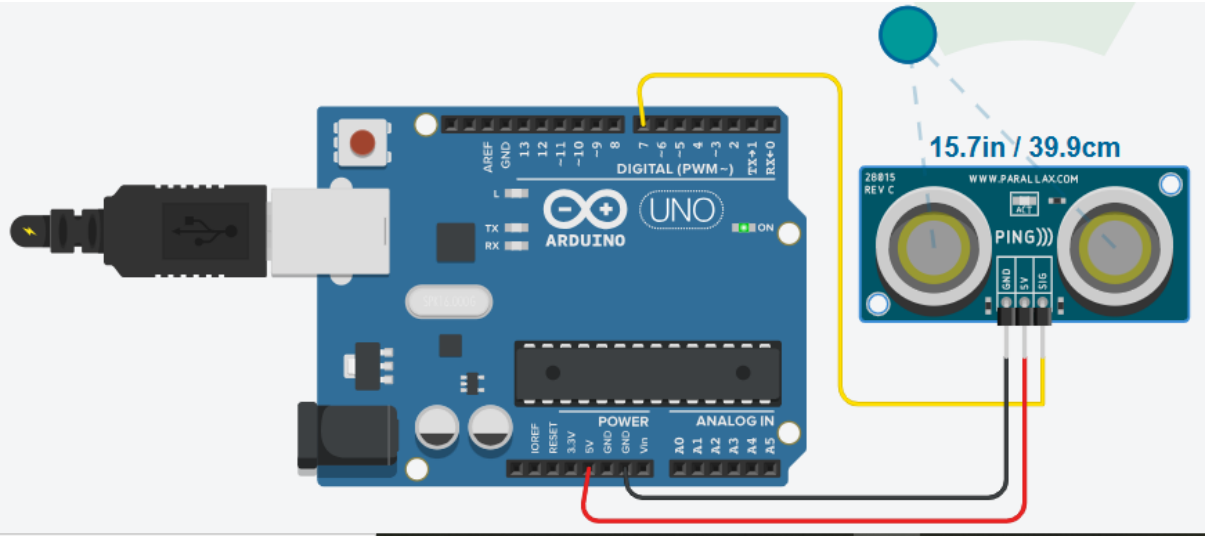
  Serial.print("Distance: ");
  Serial.print(distanceCM);
  Serial.println(" cm");

  delay(1000);
}
```

STEP 5: Start Simulation

- Click **“Start Simulation”**
- Open the **Serial Monitor** to see distance readings update continuously
- You can manually move the sensor or objects in Tinkercad to see changes in distance values.

Output:



Distance: 332.03 cm
Distance: 332.04 cm
Distance: 332.04 cm
Distance: 332.06 cm
Distance: 332.04 cm
Distance: 332.04 cm
Distance: 332.04 cm
Distance: 161.04 cmDistance: 161.02 cm
Distance: 332.08 cm
Distance: 332.08 cm

Program12:
Develop a program to simulate interfacing with the keypad module to record the keystrokes.

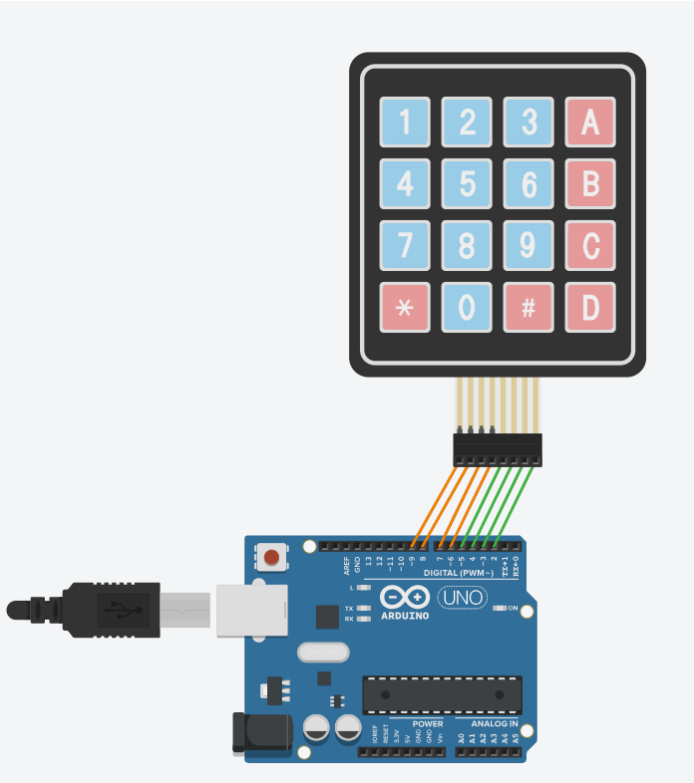
AIM

To develop a program to simulate interfacing with a keypad module and record the keystrokes using an Arduino.

Components:

Name	Quantity	Component
KEYPAD1	1	Keypad 4x4
U1	1	Arduino Uno R3

Circuit:



PROCEDURE:

STEP 1: Open Tinkercad

1. Go to https://www.tinkercad.com
2. Log in → Go to **Circuits** → Click **Create New Circuit**

STEP 2: Add Components

- Drag **Arduino Uno**
- Drag **4x4 Matrix Keypad**

STEP 3: Wiring the Keypad

Keypad Pin	Arduino Pin
1 (Row 1)	9
2 (Row 2)	8
3 (Row 3)	7
4 (Row 4)	6
5 (Col 1)	5
6 (Col 2)	4
7 (Col 3)	3
8 (Col 4)	2

STEP 4: Add the Arduino Code

Click **"Code" → Select "Text" mode** → Paste the following code:

Program Code:

```
#include <Keypad.h>

const byte ROWS = 4; // Four rows
const byte COLS = 4; // Four columns
```

```
char keys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'}
};

byte rowPins[ROWS] = {9, 8, 7, 6}; // Connect to the row pinouts of the keypad
byte colPins[COLS] = {5, 4, 3, 2}; // Connect to the column pinouts

Keypad keypad = Keypad(makeKeymap(keys), rowPins, colPins, ROWS,
COLS);

String inputBuffer = "";

void setup(){
  Serial.begin(9600);
  Serial.println("Keypad Ready. Press any key...");
}

void loop(){
  char key = keypad.getKey();
  if (key){
    inputBuffer += key;
    Serial.print("Key Pressed: ");
    Serial.println(key);
    Serial.print("Current Input: ");
    Serial.println(inputBuffer);
  }
}
```

STEP 5: Start Simulation

- Click **“Start Simulation”**
- Open the **Serial Monitor**
- Press buttons on the virtual keypad and watch the pressed keys appear on the Serial Monitor.

Output:

Keypad Ready. Press any key...
Key Pressed: 1
Current Input: 1
Key Pressed: A
Current Input: 1A
Key Pressed: 5
Current Input: 1A5
Key Pressed: #
Current Input: 1A5#
Key Pressed: D
Current Input: 1A5#D
Key Pressed: *
Current Input: 1A5#D*